Refraction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect a change in wave speed to wavelength and direction.

SUCCESS CRITERIA - I can...

1. Measure ray angles from the normal.
2. Explain frequency continuity at a stationary boundary.
3. Predict bending toward or away from the normal.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A wave snapshot has crests at **0.1**, **0.4**, **0.7 m**. **Find wavelength**. Its period is **0.05 s**; **find frequency**.

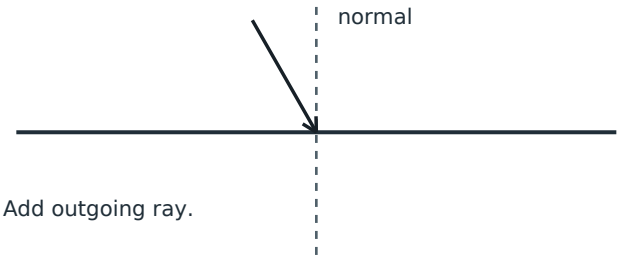
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $n_1 \sin\theta_1 = n_2 \sin\theta_2$; $n = c/v$; $v = f\lambda$.
Angles from the normal. Stationary boundary: frequency stays constant.

Worked reasoning: Light enters a slower medium at an angle to the normal and bends toward the normal. The normal is perpendicular to the boundary.

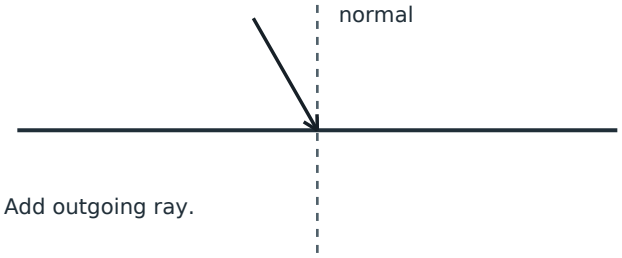
Your turn: Draw and **label the boundary**, normal, incident ray and refracted ray.



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

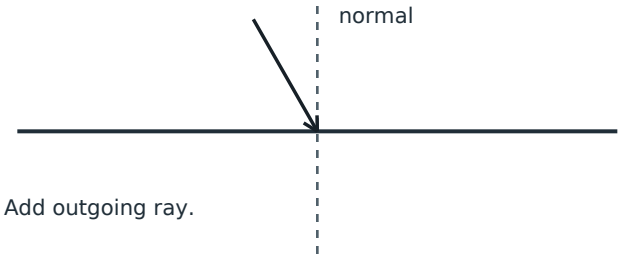
Mark the normal at the boundary. A ray enters a slower medium at an angle. **Sketch the refracted direction**.



Marzano check - circle: ① ② ③

04 Apply the physics

A laser ray travels from air (n = **1.00**) into glass (n = **1.50**) at **30°** to the normal. **Label both media** and angles, then use $n_1 \sin\theta_1 = n_2 \sin\theta_2$ to **find the refracted angle**.



Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

A student measures an incidence angle from the surface instead of the normal. If surface angle is **20°**, **what angle** belongs in Snell's law?

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A **0.4 kg** metal cools from **92** to **32°C** while **0.3 kg** water warms from **12** to **32°C**. Use water c=**4200**; neglect cup/losses. **Find metal c**.
Label temperatures, samples and relevant energy transfers.

Sample A

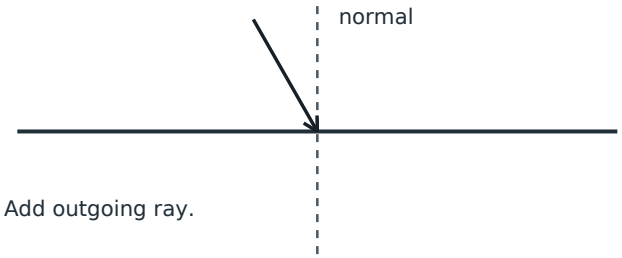
Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: ① ② ③

07 Exit check

From glass (n=**1.5**) to air, find critical angle using sin θ_c=**1/1.5**. **State what happens above** that angle in this ideal case.

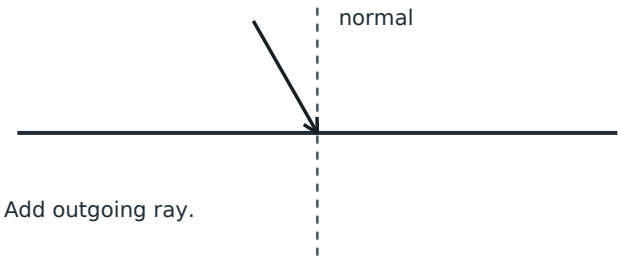


Add outgoing ray.

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

A ray travels from water (n = **1.33**) into glass (n = **1.50**) at **40°** to the normal. Use $n_1 \sin\theta_1 = n_2 \sin\theta_2$ to **find the refracted angle**. Then **find the critical angle** for travel from this glass into water. **Label both cases**.



Add outgoing ray.

Marzano check - circle: ① ② ③